

# Skills Summary

## Solving Linear Systems by Substitution

*Use this reference while completing your worksheet or when studying.*

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### 1. One variable is already isolated

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**The method:** substitution replaces one variable by an *equal expression*, turning two equations in two unknowns into **one** equation in one unknown.

(1) Find the equation solved for a variable.      (2) Put that expression — in parentheses — into the *other* equation.

(3) Solve for the one remaining variable.      (4) Back-substitute to get the partner.      (5) Check in the untouched equation.

**Example:**  $y = x + 2$  and  $2x + y = 11$

**Step 1.** The first equation already says what  $y$  equals, so substituting it into the second leaves an equation in  $x$  alone — no isolating work needed.

**Step 2.**  $2x + (x + 2) = 11 \Rightarrow 3x + 2 = 11 \Rightarrow x = 3$ ; then  $y = 3 + 2 = 5$ . Check:  $2(3) + 5 = 11$   
✓ Solution: **(3, 5)**

**Try it:**  $y = 2x - 1$  and  $3x + y = 19$

check: **(4, 7)**

### 2. Isolate a variable first (isolate-then-substitute)

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**Rule:** when nothing is isolated, isolate the variable whose coefficient is 1 or  $-1$  — that choice keeps fractions out of the substitution.

From  $x + y = 9$ :  $y = 9 - x$ . Substituting a subtraction needs parentheses:  $2x - (9 - x) = 2x - 9 + x$ .

**Example:**  $x + y = 9$  and  $2x - y = 3$

**Step 1.** Neither equation is solved for a variable, and  $y$  in the first has coefficient 1, so isolating there is the fraction-free choice.

**Step 2.**  $y = 9 - x$ , so  $2x - (9 - x) = 3 \Rightarrow 3x - 9 = 3 \Rightarrow x = 4$ ; then  $y = 9 - 4 = 5$ . Check:  $2(4) - 5 = 3$  ✓ Solution: **(4, 5)**

**Try it:**  $x + 2y = 11$  and  $3x - y = 5$

check: **(3, 4)**

### 3. When both variables disappear (special-case)

**Read what is left behind:**

$-6 = 1$  or  $12 = 5$  (**false**)  $\Rightarrow$  **no solution**; the lines are parallel.

$8 = 8$  or  $0 = 0$  (**true**)  $\Rightarrow$  **infinitely many solutions**; the two equations describe the same line.

Losing the variables is information, not a mistake — never write “ $x = 0$ ”.

**Example:**  $2x + y = 6$  and  $4x + 2y = 5$

**Step 1.** Nothing is isolated, so isolate  $y = 6 - 2x$  and substitute — then read whatever statement survives.

**Step 2.**  $4x + 2(6 - 2x) = 5 \Rightarrow 4x + 12 - 4x = 5 \Rightarrow 12 = 5$ , which is false. No pair can work: no solution (parallel lines).

**Try it:**  $y = 3x - 4$  and  $6x - 2y = 8$ . Substitute, say what the surviving statement means, then give one pair that works.

(2, 2) check: infinitely many; e.g.

### 4. Building the system from a story (word-model)

**Two unknowns need two equations.** Name both unknowns with units, then write one equation per *kind* of total in the story:

a **count** equation (how many things) and a **value** equation (money, acid, length). Solve the count equation for a variable — its coefficients are 1 — and substitute into the value equation.

**Example:** 12 coins, dimes and quarters, worth 255 cents. How many of each?

**Step 1.** Let  $x$  be dimes and  $y$  quarters: the count total gives one equation and the money total the other, so substitution starts from the count equation.

**Step 2.**  $x + y = 12 \Rightarrow x = 12 - y$ ; then  $10(12 - y) + 25y = 255 \Rightarrow 120 + 15y = 255 \Rightarrow y = 9$ , so  $x = 3$ . Dimes and quarters: **(3, 9)**

**Try it:** 15 tickets, adult at 6 dollars and student at 4 dollars, worth 76 dollars. Find  $(x, y)$  with  $x$  adult tickets.

(2, 8) check: